

Active Thermal Control & Ammonia Waste Heat Radiator Loops

1. Dual-Loop Thermal Architecture

Internal water coolant loops absorb electronics and human metabolic heat inside the habitat, transferring thermal energy via plate-and-frame heat exchangers to an external anhydrous ammonia loop.

2. Deployable Radiator Panels

Composite aluminum-honeycomb radiator wings radiate heat into deep space at night (effective sink temperature 180 K) with emissivity of 0.91 and solar absorptivity of 0.08.

3. Passive Freeze Protection

Proportional bypass valves throttle ammonia flow during extreme Martian winter nights (-120 C) to prevent coolant freezing, maintaining core loop return temperatures above -35 C.